

Autoencoder

• Goal:

Learn a compact (latent) representation of data that can reconstruct the original input.

• Encoder:

Maps input to latent spaces:

$$h = f(w_e x + b_e)$$

- x : input vector
- w_e, b_e : encoder weights and bias
- $f(\cdot)$: activation (ReLU | tanh)
- h : latent (compressed) vector

• Decoder:

Reconstructs inputs from latent vector:

$$\hat{x} = g(w_d h + b_d)$$

- w_d, b_d : decoder weights and bias
- $\hat{x}$: reconstructed input

• Loss Function (Reconstruction Error)

Usually Mean Squared Error (MSE):

$$L = \frac{1}{n} \sum (x - \hat{x})^2$$

Measures how close reconstruction is to original data.

• Training

- Minimize reconstruction loss
- Use gradient descent
- Update all weights via backpropagation.

• Latent Space

- Lower dimensional than input
- Captures essential structure
- Similar inputs $\rightarrow$ nearby latent points

• Conclusion

Autoencoders learn useful data representations by compressing and reconstructing inputs without labels.